SENIOR SCHOOL · GRADE 10 · KENYA

CHEMISTRY UNPACKED

A learner's pamphlet built around the KICD Grade 10 Chemistry curriculum — the atom, the periodic table, bonding, periodicity, acids, bases and salts, explained the way you'll actually use them: in the lab, in the kitchen, and on the farm.

1	2	3	4	5	6
H	**PT**	**B**	**Pd**	**AB**	**Sa**
ATOM	TABLE	BONDS	PERIODS	ACIDS	SALTS

AGES 14–16

01

In

INTRO

TOPIC 01 · INTRODUCTION TO CHEMISTRY

Why Chemistry Matters

Key inquiry: How is the study of Chemistry important in our society?

WHAT IS CHEMISTRY?

Chemistry is the branch of science that studies what things are made of, how they behave, and how they change into new substances. It sits between Biology and Physics, and it explains everything from why milk turns sour to why a matchstick catches fire.

BRANCHES OF CHEMISTRY

- **Inorganic Chemistry** — elements, atoms, and compounds without carbon chains.
- **Organic Chemistry** — carbon-based compounds like fuels and plastics.
- **Physical Chemistry** — energy, reactions and how fast/why they happen.
- **Analytical Chemistry** — identifying and measuring substances.
- **Biochemistry** — chemistry inside living things.

CHEMISTRY IN YOUR DAY

Cooking ugali (starch breaking down with heat), charging a phone (electrochemical cells), treating water with chlorine, making soap from fat and lye, and manufacturing fertiliser for maize — all chemistry, all Kenyan, all daily.

CAREERS TO EXPLORE

Pharmacy, medicine, agriculture, food technology, forensic science, petroleum and mining, environmental science, and manufacturing engineering all build directly on Chemistry.

Notice: these careers are open to every learner, regardless of gender

SAFETY FIRST: DRUGS & SUBSTANCES

Every learner has a right to a safe, healthy learning environment. Know the difference between a prescribed dose (given by a doctor for a reason) and substance misuse, and understand why reading product labels — expiry dates, dosage, cautions — protects you as a consumer.

02

At

ATOM

TOPIC 02 · THE ATOM

Inside the Atom

Key inquiry: How are electrons arranged in an atom?

STRUCTURE AT A GLANCE

- **Protons** (+) and **neutrons** live in the tiny, dense **nucleus**.
- **Electrons** (−) move around the nucleus in **energy levels** (shells), and within these, in regions called **orbitals** — **s** and **p**.
- **Atomic number (Z)** = number of protons = number of electrons in a neutral atom.
- **Mass number (A)** = protons + neutrons.

Z	**A−Z**	**Z**
PROTONS	NEUTRONS	ELECTRONS

ISOTOPES & RELATIVE ATOMIC MASS

Isotopes are atoms of the same element with the same number of protons but a different number of neutrons. Relative atomic mass is the weighted average of an element's isotopes, based on how abundant each one is in nature.

FILLING ORDER (FIRST 20 ELEMENTS)

1s → 2s → 2p → 3s → 3p → 4s

Example — Sodium (Na, Z = 11): $1s^2\ 2s^2\ 2p^6\ 3s^1$

FROM DALTON TO RUTHERFORD

Dalton (1803): atoms are solid, indivisible spheres. **Rutherford's gold foil experiment (1909):** most alpha particles passed straight through gold foil, but a few bounced back — proving atoms are mostly empty space with a small, dense, positive nucleus.

TRY THIS

Model an atom using beads, bottle tops or seeds of different sizes/colours for protons, neutrons and electrons — a hands-on way to see energy levels filling up.

03

Pt

TABLE

TOPIC 03 · THE PERIODIC TABLE

Reading the Periodic Table

Key inquiry: Why is the study of the periodic table important?

GROUPS & PERIODS

Elements are arranged by increasing atomic number. A **group** (vertical column) shares the same number of valence electrons; a **period** (horizontal row) shares the same number of occupied energy levels. Elements in the same group behave alike — that's what makes the table so powerful for prediction.

- Group I — Alkali metals
- Group II — Alkaline earth metals
- Group VII — Halogens
- Group VIII/0 — Noble gases

ION FORMATION

Atoms gain or lose electrons to reach a stable, noble-gas electron arrangement. Losing electrons forms a positive ion (**cation**); gaining electrons forms a negative ion (**anion**). The number of electrons lost/gained relates directly to **valency** and **oxidation number**.

WRITING FORMULAE

Cross the valencies (or use oxidation numbers) to write a compound's formula. Example: Magnesium (2+) and Chloride (1−) → **$MgCl_2$**. Common radicals to know: OH^-, NO_3^-, SO_4^{2-}, CO_3^{2-}, NH_4^+.

BALANCING EQUATIONS

A balanced equation has the same number of each type of atom on both sides — it obeys the Law of Conservation of Mass. Example:

$$2Mg + O_2 \rightarrow 2MgO$$

WHY IT MATTERS

Predicting an unknown element's properties from its position in the table is exactly how new elements were discovered — and how chemists today predict how a substance will react before ever testing it in the lab.

04

Bd

BONDS

TOPIC 04 · CHEMICAL BONDING

What Holds Matter Together

Key inquiry: How do chemical bonds influence the properties of substances?

TYPES OF BONDS

- **Ionic** — electrons transferred between a metal and a non-metal (e.g. NaCl).
- **Covalent** — electrons shared between non-metals (e.g. H_2O, CO_2).
- **Dative (coordinate) covalent** — both shared electrons come from one atom (e.g. NH_4^+).
- **Metallic** — a "sea" of delocalised electrons around metal ions.
- **Hydrogen bonding & Van der Waals** — weaker forces between molecules.

STRUCTURE EXPLAINS PROPERTIES

Giant ionic structures (NaCl) have high melting points and conduct electricity only when molten or dissolved. Giant covalent structures (diamond, SiO_2) are extremely hard with very high melting points. Simple molecular substances (I_2, CO_2) melt/boil at low temperatures. Metals conduct electricity and are malleable due to delocalised electrons.

DIAMOND VS GRAPHITE

Both are pure carbon — same element, different bonding arrangement. Diamond's rigid 3-D lattice makes it the hardest natural material. Graphite's layered structure lets sheets slide over each other, which is why it's soft enough to write with and conducts electricity.

PROJECT IDEA

Build a model of NaCl, SiO_2, graphite or diamond bonding using local materials — beads, sticks, bottle tops, or clay — and explain how the structure connects to the real-world use of that substance.

EVERYDAY LINK

Aluminium's metallic bonding makes it light, strong and an excellent electrical conductor — which is why it's used in overhead power cables across Kenya.

05

Pd

PERIODS

TOPIC 05 · PERIODICITY

Trends Across the Table

Key inquiry: How does the position of an element influence its properties?

TRENDS ACROSS PERIOD 3

- Atomic size **decreases** left to right (more protons pull electrons in tighter).
- Ionisation energy generally **increases** left to right.
- Elements shift from metallic (Na, Mg — good conductors, malleable) to non-metallic (S, Cl — poor conductors, brittle solids or gases).

GROUP I & II: THE REACTIVE METALS

Reactivity **increases down** Group I and II. Sodium and potassium react vigorously with cold water, fizzing and sometimes catching fire, releasing hydrogen gas and forming an alkaline hydroxide solution.

GROUP VII: THE HALOGENS

Chlorine (pale green gas), bromine (red-brown liquid) and iodine (grey solid, purple vapour) get darker and less reactive down the group. Chlorine displaces bromine and iodine from their salt solutions — a classic reactivity test.

WHY KENYA CARES

Chlorine's bleaching and disinfecting action is exactly why it's used to treat drinking water and swimming pools across the country — a direct, life-saving application of Group VII chemistry.

SAFETY NOTE

Chlorine gas is toxic. Always prepare and test it in a fume cupboard or well-ventilated space, following your teacher's safety instructions closely.

06

Ab

ACIDS

TOPIC 06 · ACIDS AND BASES

Sour, Bitter & Everything Between

Key inquiry: What common household substances are acidic and/or basic?

WHAT ACIDS AND BASES DO

- Acid + metal → salt + hydrogen gas
- Acid + metal carbonate → salt + water + carbon dioxide
- Acid + metal oxide/hydroxide → salt + water (neutralisation)

Some oxides/hydroxides react with both acids and bases — these are **amphoteric**.

STRONG VS WEAK

Strength is about how fully a substance dissociates (splits into ions) in water — not the same as concentration. Strong acids (e.g. hydrochloric acid) dissociate almost completely and conduct electricity well. Weak acids (e.g. ethanoic acid) dissociate only partly.

THE PH SCALE

0 acidic 7 neutral 14 basic

A universal indicator turns a range of colours to show exactly where a substance sits on this scale.

IN YOUR KITCHEN & SHAMBA

Lemon juice and vinegar (acidic), soap and baking soda solution (basic), and ammonia-based cleaners (basic) are all around you. Farmers use lime (a base) to correct acidic soil before planting.

07

Sa

SALTS

TOPIC 07 · INTRODUCTION TO SALTS

Salts: More Than Table Salt

Key inquiry: How can salts be prepared, and how do they behave in air?

TYPES OF SALTS

- **Normal salt** — all replaceable hydrogens of the acid replaced (NaCl).
- **Acidic salt** — only some hydrogens replaced ($NaHCO_3$).
- **Basic salt** — contains a hydroxide/oxide ion along with the salt.
- **Double salt** — two salts crystallising together (e.g. potash alum).

PREPARING SALTS

Method depends on solubility: soluble salts are made by reacting acid with an excess metal, base or carbonate, then evaporating; insoluble salts are made by **precipitation** — mixing two soluble solutions so an insoluble salt forms directly.

BEHAVIOUR IN AIR

Hygroscopic salts absorb moisture from air without dissolving. **Deliquescent** salts absorb so much moisture they dissolve into a solution. **Efflorescent** salts lose their water of crystallisation to the air and crumble to powder.

SALTS THAT FEED AND HEAL

Inorganic fertilisers (ammonium and phosphate salts) boost crop yields but, if overused, can cause water pollution and eutrophication in rivers and lakes. Salts also appear in oral rehydration solutions, table salt, and food preservation.

LAB SAFETY

Always dispose of laboratory waste responsibly after preparing salts — it's both a safety habit and an environmental one.

Qz
RECAP

QUICK RECAP

Test Yourself

Cover the pamphlet and see how many you can answer.

QUIZ

1. What determines an atom's atomic number?
2. Why do elements in the same group behave similarly?
3. Name the bond type in sodium chloride, and in oxygen gas (O_2).
4. Why does reactivity increase down Group I but down Group VII it decreases?
5. What pH range would you expect for vinegar? For a soap solution?
6. Give one method for preparing an insoluble salt.
7. What's the difference between a hygroscopic and a deliquescent salt?

CAREERS BUILT ON THIS PAMPHLET

- Pharmacy
- Medicine
- Agriculture
- Food science
- Forensics
- Mining & petroleum
- Environmental science
- Chemical engineering

KEEP EXPLORING

Chemistry rewards curiosity. Ask "what is this made of?" and "why does it behave this way?" about the world around you — your shamba, your kitchen, your local factory — and you're already doing science.